Results: Subfield Structure

Results: Subfield Structure I

The subfield taxonomy is defined entirely in configuration; for this instance it spans 6 buckets (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience). Each record is assigned to the highest-priority bucket whose keywords it matches, so the distribution reflects the configured taxonomy rather than a fixed schema. Table 2 (previous section) reports the counts; the largest bucket is **Clinical Sleep** (63.0%).

Per-Subfield Characterization I

The subfield breakdown reveals the multi-disciplinary nature of the modafinil literature:

- **Clinical Sleep** dominates at 63.0%, reflecting the drug's primary indication for narcolepsy, shift-work disorder, and obstructive sleep apnea. This bucket includes randomized controlled trials, meta-analyses of efficacy, and long-term safety studies in sleep-disorder populations.
- **Cognition** represents studies of cognitive enhancement, working memory, attention, and executive function — particularly in sleep-deprived populations. This subfield has grown with the broader interest in neuroenhancement and "smart drugs."
- **Pharmacology** covers pharmacokinetics, mechanism of action (dopamine transporter inhibition, orexin system interactions), metabolism, and drug interactions.

Per-Subfield Characterization II

- **Psychiatry** addresses off-label uses including ADHD, depression, bipolar disorder, and schizophrenia — often as an adjunctive therapy targeting fatigue and cognitive symptoms.
- **Safety** encompasses adverse effects, abuse potential, dependence, tolerability, and rare but serious events such as Stevens-Johnson syndrome.
- **Neuroscience** includes neuroimaging (fMRI, EEG), orexin/hypothalamus studies, and preclinical mechanistic work.

Because the taxonomy is data, not code, re-targeting the template to another topic — or refining the buckets for the same topic — changes this section's structure and numbers without any change to the analysis code. The subfield assignment also feeds the temporal and citation analyses, allowing per-subfield growth and connectivity to be read off the same artifacts.